

data_capstone description.

Penguin Species Biometric Analysis

Welcome to the Penguin Species Biometric Analysis repository! This project involves the analysis and visualization of penguin biometric data from the Palmer Archipelago in Antarctica. The dataset includes various measurements of penguins, including species, culmen length, culmen depth, and flipper length.

Project Overview

In this project, we will:

- Load and clean the dataset to prepare it for analysis.
- Standardize column names for consistency.
- Visualize the distribution of different penguin species.
- Explore relationships between various biometric measures, such as culmen length and depth.
- Analyze the distribution of flipper length by species.
- Create a correlation heatmap to examine the relationships between numeric variables.

Features

1. **Data Cleaning:** Rows with missing values are removed, and column names are standardized for better accessibility.
2. **Data Visualizations:**
 - Distribution of penguin species using a count plot.
 - Scatter plot to show the relationship between culmen length and culmen depth with color coding based on species.
 - Box plot displaying the distribution of flipper length across different species.
 - Correlation heatmap demonstrating the correlations between various biometric measures.

Technologies Used

- Python
- Pandas
- Seaborn
- Matplotlib

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Visualization Examples

Visualizations generated in this analysis include:

- Count plot showing the number of individuals per species.
- Scatter plot illustrating the relationship between culmen length and culmen depth.
- Box plot for flipper length distribution by species.
- A heatmap revealing the correlation between different numeric measures.